

Advanced Telecommunication Technologies and Management

1



MASTERS 2

**FACULTY OF ECONOMIC SCIENCES AND BUSINESS ADMINISTRATION
BUSINESS INFORMATICS DEPARTMENT
LEBANESE UNIVERSITY**

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Course objectives

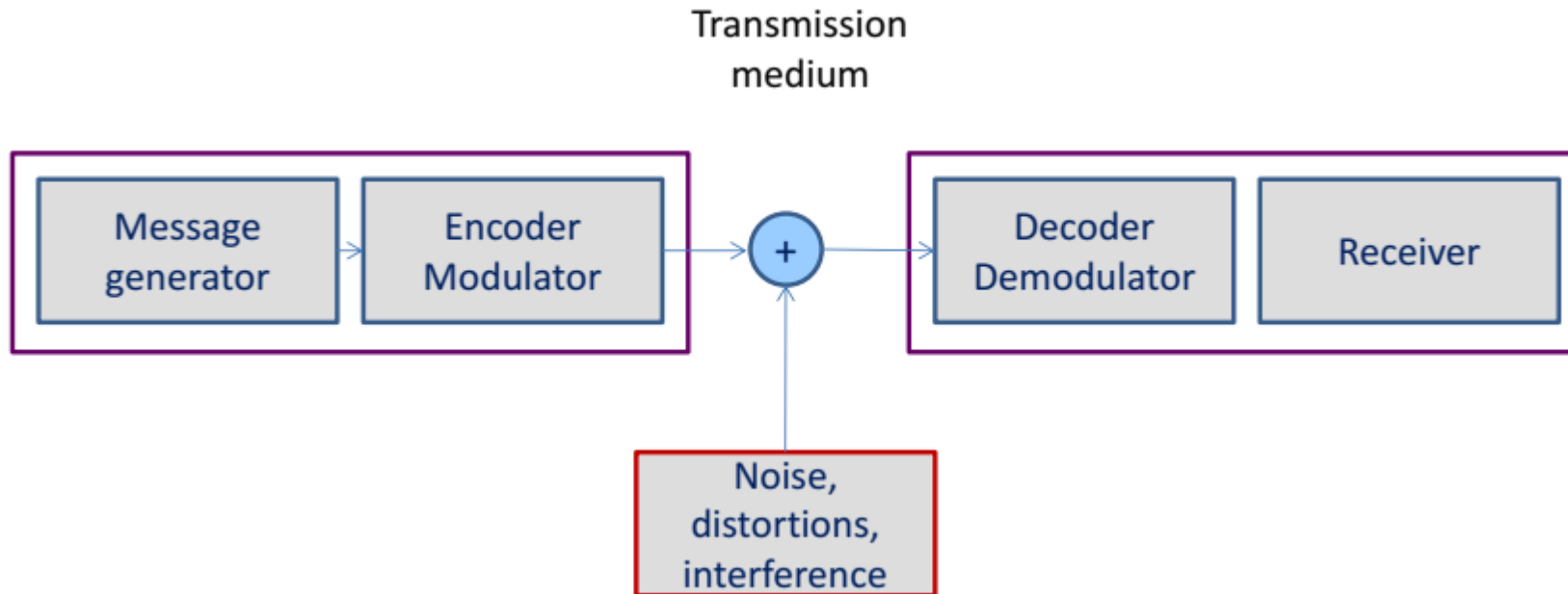
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- Computer Communication Networks
- **Signal Theory**
- Digital Communications
- Telecommunication Technologies



Signal Theory

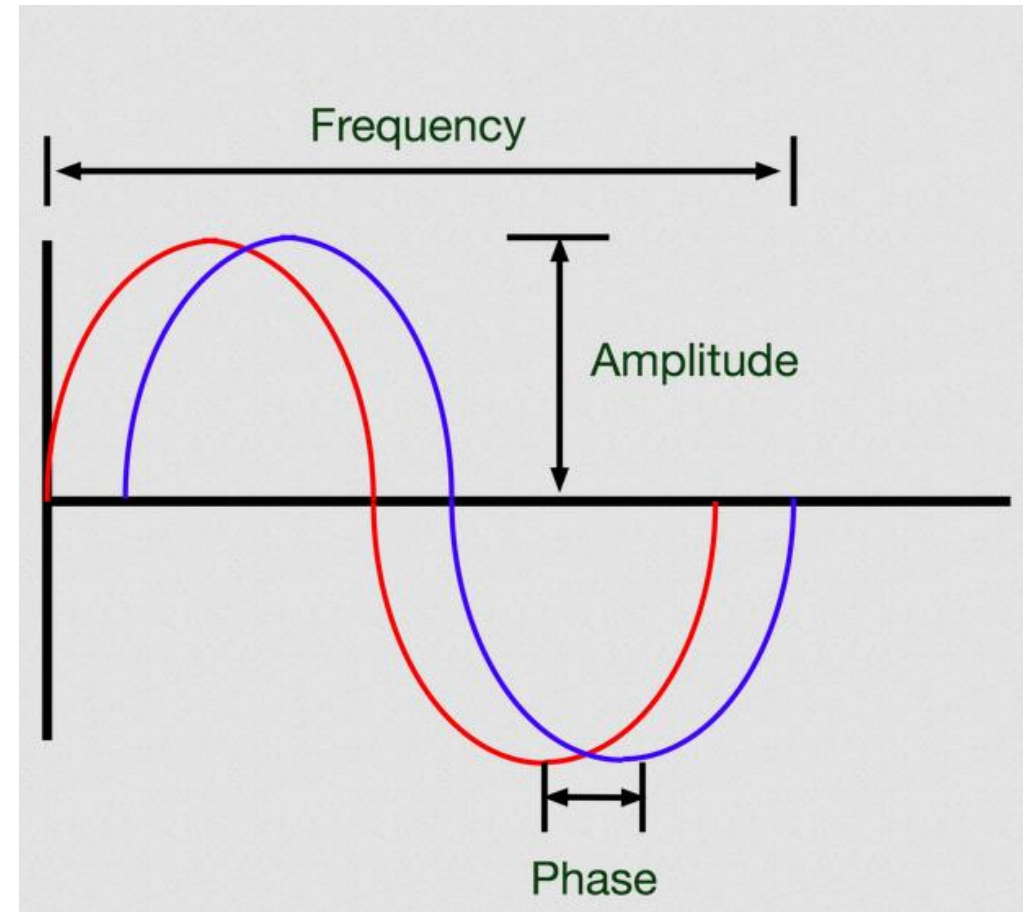
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Signal Characteristics

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- Amplitude A
- Frequency f
 - Pulsation $\omega = 2\pi f$
- Phase φ

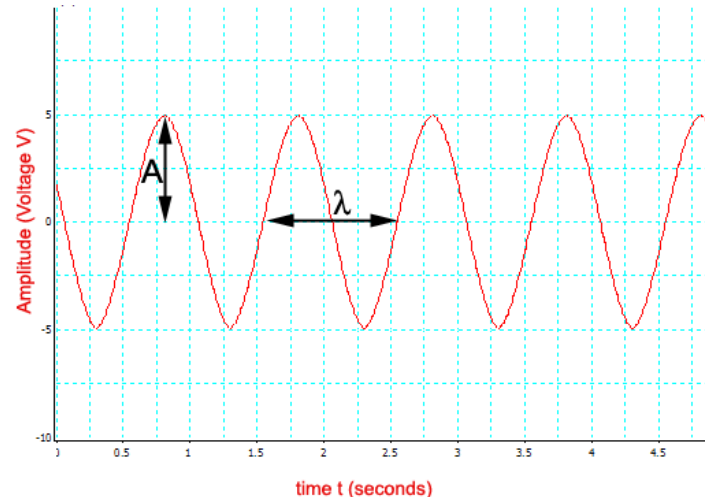


Signal Classification

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- Deterministic Signals

- $s(t) = A_0 \cos(\omega_0 t + \varphi)$
- A_0 , ω_0 and φ are constants, (purely deterministic signal).
 - ✦ When the amplitude, pulsation and phase shift have known values, it is perfectly possible to compute the value that the signal will obtain at any freely chosen time instant t .

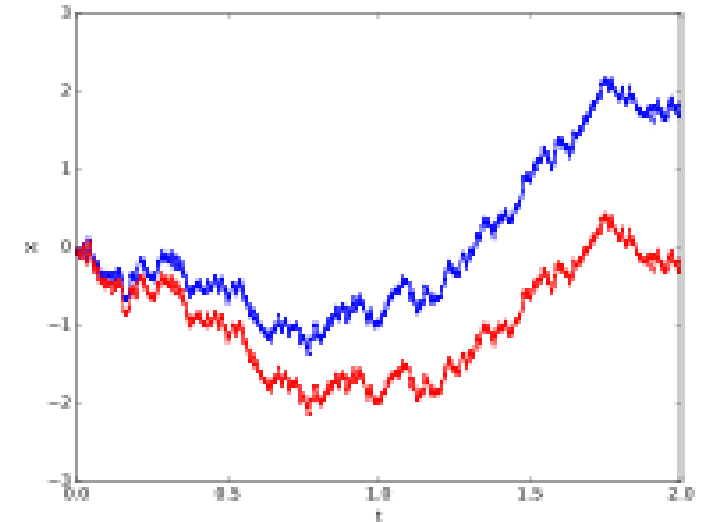


Signal Classification

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- Stochastic Signals

- The values that the signal will gain at any time instance cannot be computed from a deterministic model
 - ✦ i.e. the successive values that a signal will take at any chosen time instance, is not known in advance
- Unpredictable
- Described using large sets of statistical observations.
- Treated in a statistical way.

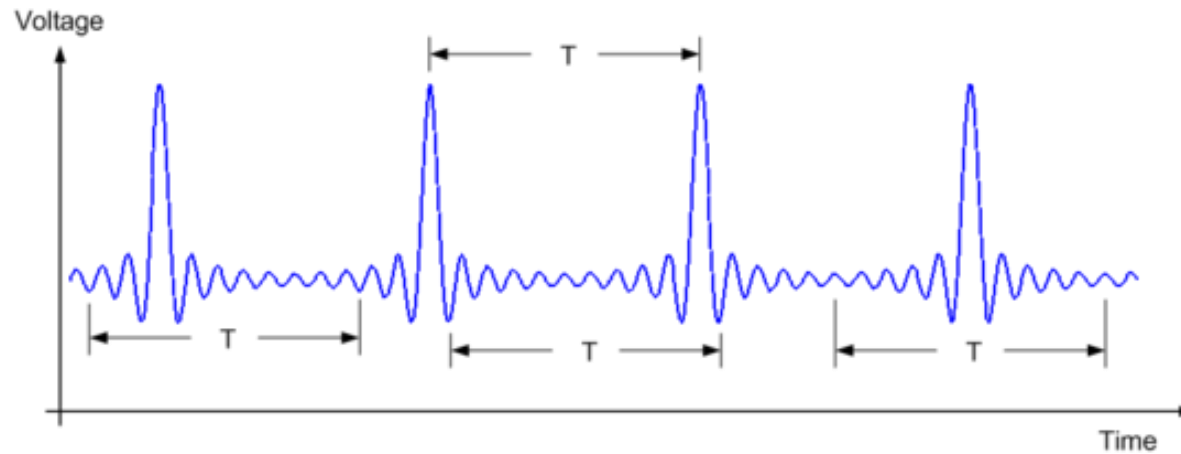


Signal Classification

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- Periodic Signals

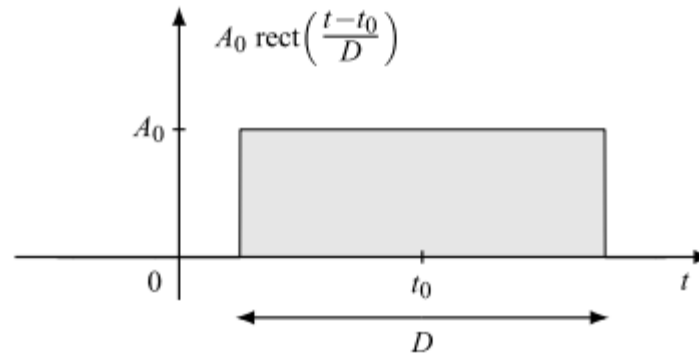
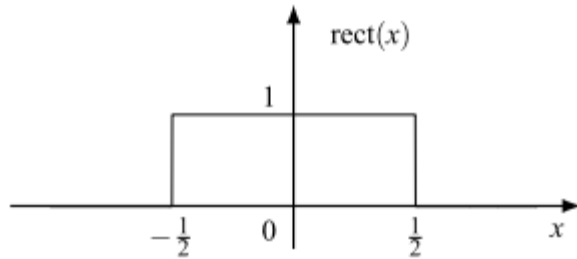
- $s(t + KT) = s(t)$



Signal Classification

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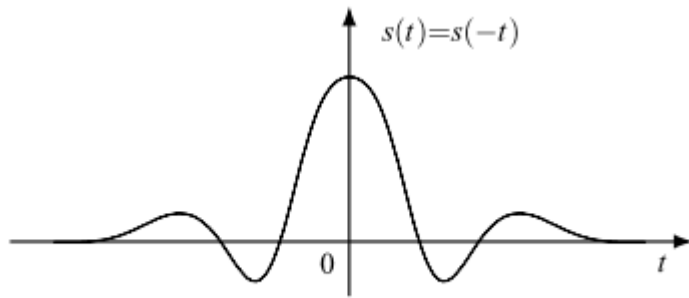
- Rectangular Signal



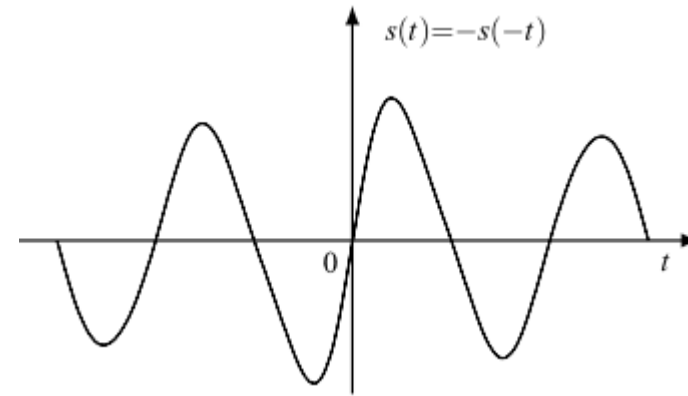
Signal Symmetries

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- Even vs Odd Signal



Even Signal



Odd Signal



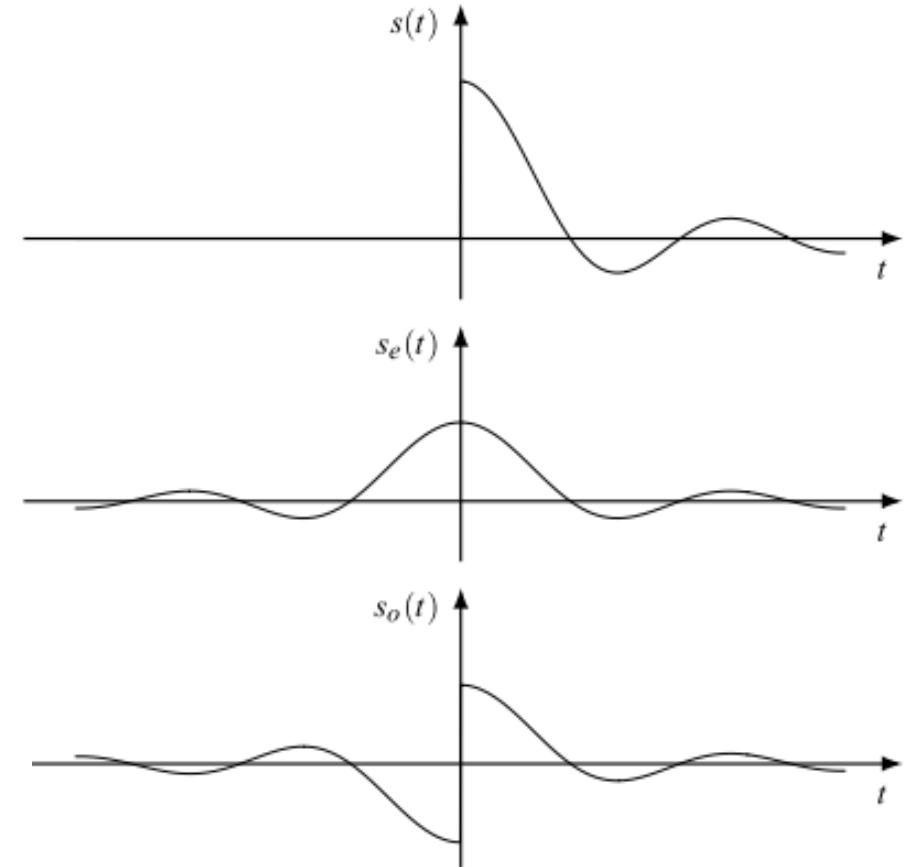
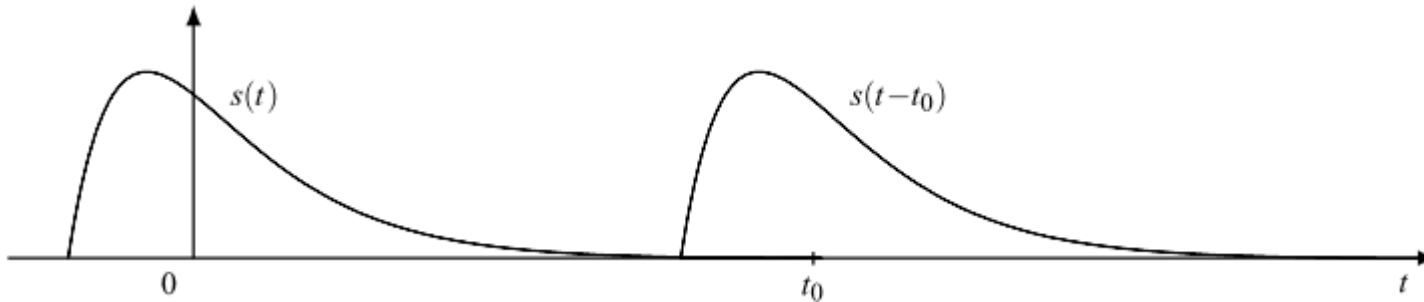
Signal Symmetries

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- Even vs Odd Signal

- $s(t) = s_e(t) + s_o(t)$

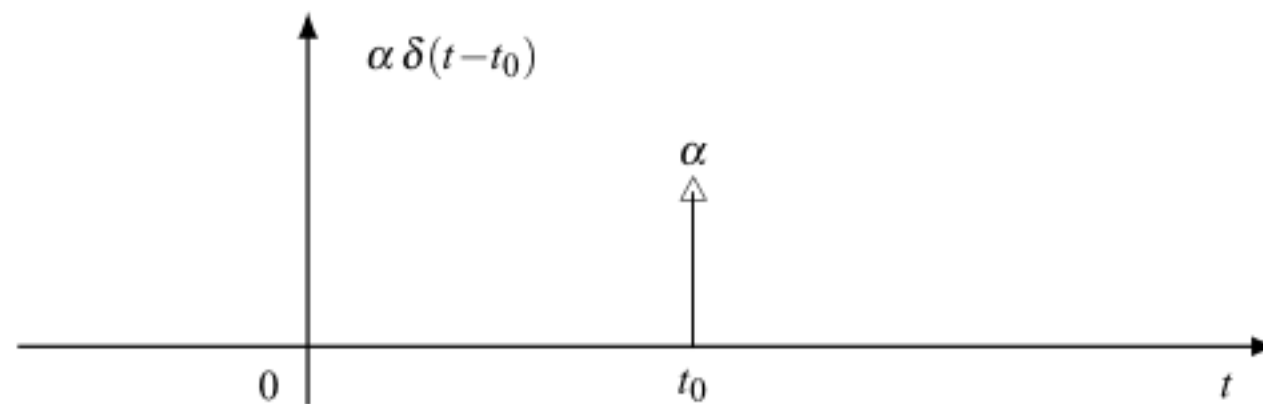
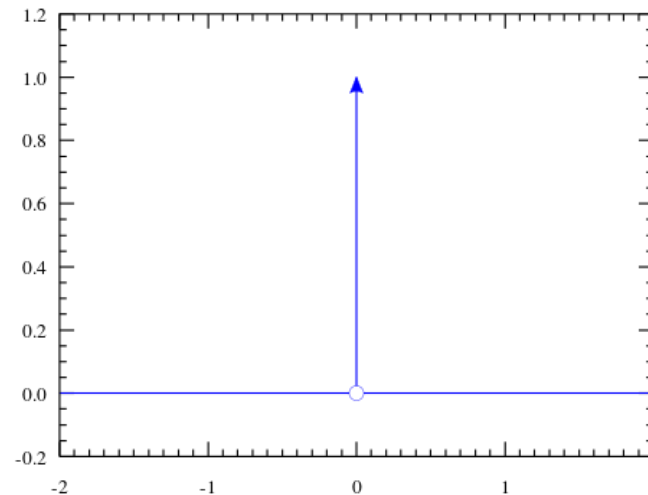
- Time shift



Impulse

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- Dirac function $\delta(t)$



Fourier Series

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- Continuous-time signals are examined in the frequency domain.
- The tool is given by the Fourier series for periodic signals
- In 1822, Joseph Fourier proved that an arbitrary (real) function of a real variable $s(t)$, $t \in \mathbb{R}$, having period T_p , can be expressed as the sum of a series of sine and cosine functions with frequencies multiple of the fundamental frequency $F = \frac{1}{T_p}$:

$$s(t) = A_0 + \sum_{k=1}^{\infty} [A_k \cos 2\pi k F t + A_k \sin 2\pi k F t]$$



Fourier Transform

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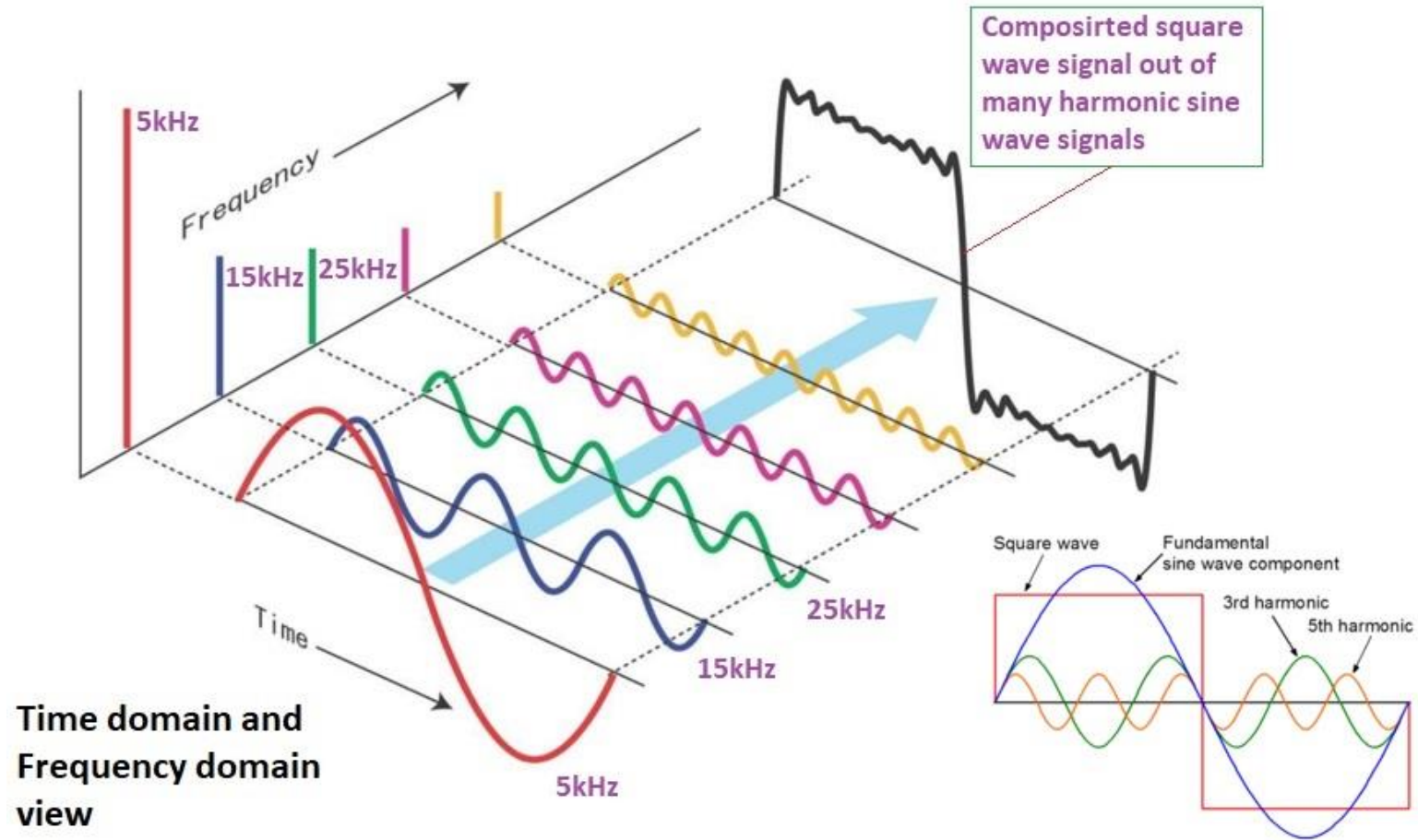
- Frequency Domain

$$S(f) = \int_{-\infty}^{+\infty} s(t) e^{-i2\pi Ft} dt$$



Fourier Transform

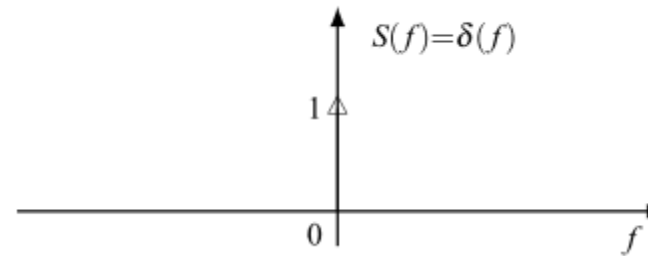
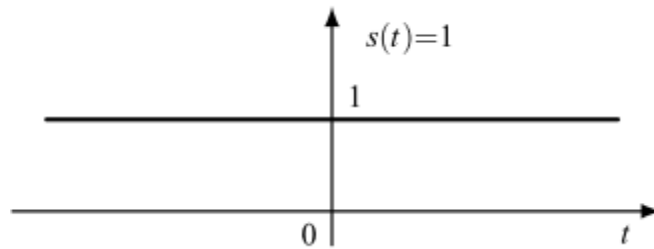
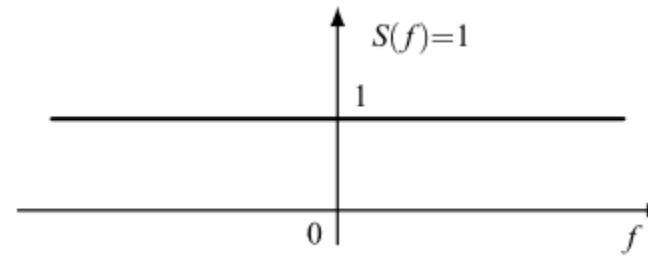
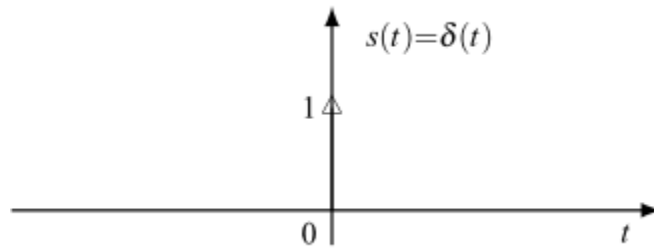
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Fourier Transform

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- Fourier transforms of the impulse and of the unit signal



Fourier Transform

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- Demo Matlab: Fast Fourier Transform

```
close ALL
```

```
clc
```

```
Fs = 2000;           % Sampling frequency
```

```
T = 1/Fs;           % Sampling period
```

```
L = 5500;           % Length of signal
```

```
t = (0:L-1)*T;       % Time vector
```



Fourier Transform

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- Demo Matlab: Fast Fourier Transform

```
%Signal containing a 50 Hz sinusoid of amplitude 0.7 and a 120 Hz  
sinusoid of amplitude 1.
```

```
s1=0.7*sin(2*pi*50*t);
```

```
s2 = sin(2*pi*120*t);
```

```
S=s1+s2;
```

```
%S = 0.7*sin(2*pi*50*t) + sin(2*pi*120*t);
```

```
subplot(1,3,1)
```

```
plot(1000*t(1:250),s1(1:250))
```

```
title('s1(t)')
```

```
xlabel('t (milliseconds)')
```

```
ylabel('s1(t)')
```



Fourier Transform

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- Demo Matlab: Fast Fourier Transform

```
subplot(1,3,2)
plot(1000*t(1:250),s2(1:250))
title('s2(t)')
xlabel('t (milliseconds)')
ylabel('s2(t)')

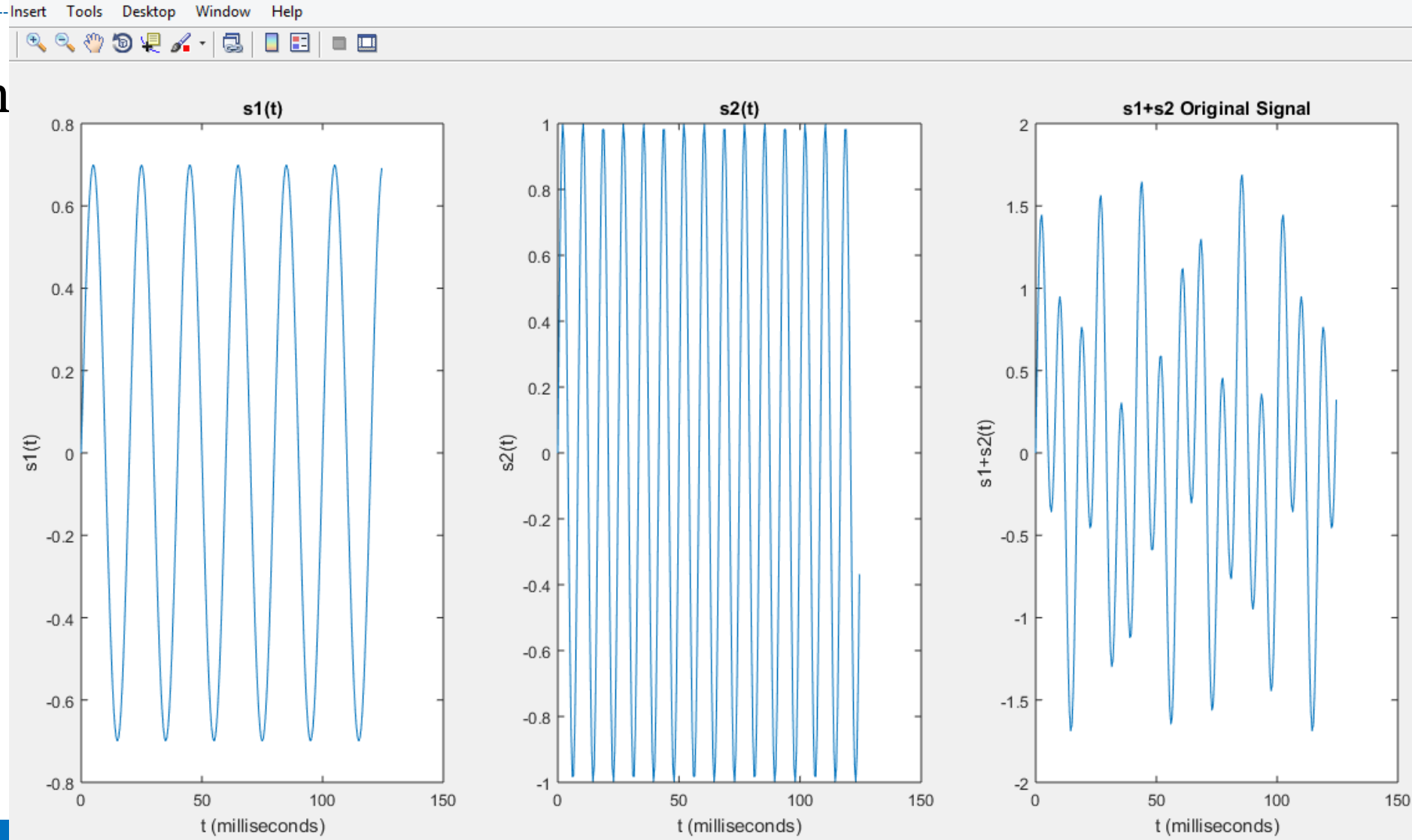
subplot(1,3,3)
plot(1000*t(1:250),S(1:250))
title('s1+s2 Original Signal')
xlabel('t (milliseconds)')
ylabel('s1+s2(t)')
```



Fourier Transform

Figure 1

- Dem



Fourier Transform

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- Demo Matlab: Fast Fourier Transform

% Fourier transform of the original, uncorrupted signal and retrieve the exact amplitudes, 0.7 and 1.0.

```
Y = fft(S);
```

```
P2 = abs(Y/L);
```

```
P1 = P2(1:L/2+1);
```

```
P1(2:end-1) = 2*P1(2:end-1);
```

```
f = Fs*(0:(L/2))/L;
```

```
plot(f,P1)
```

```
title('Single-Sided Amplitude Spectrum of S(t)')
```

```
xlabel('f (Hz)')
```

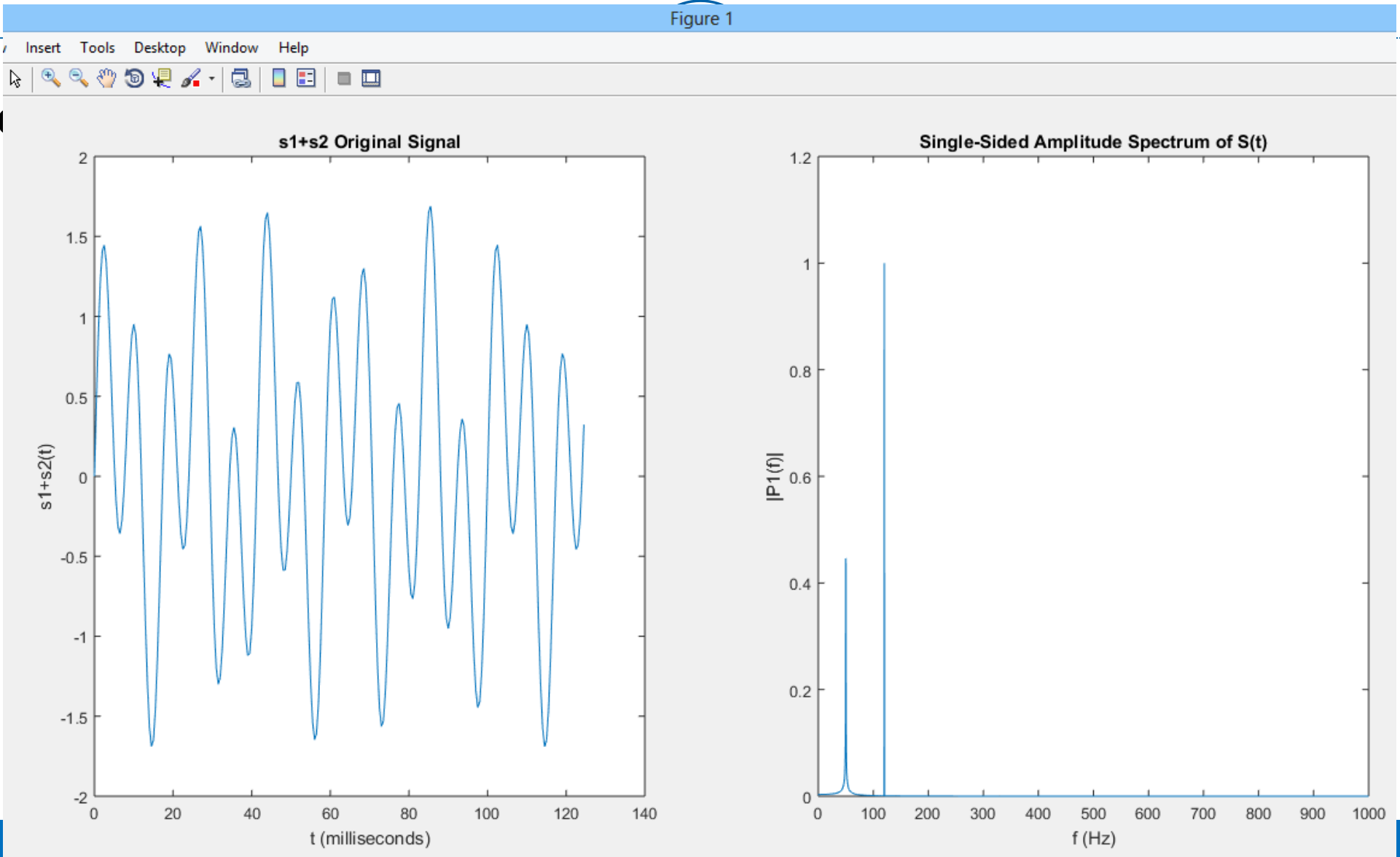
```
ylabel('|P1(f)|')
```



Fourier Transform

Figure 1

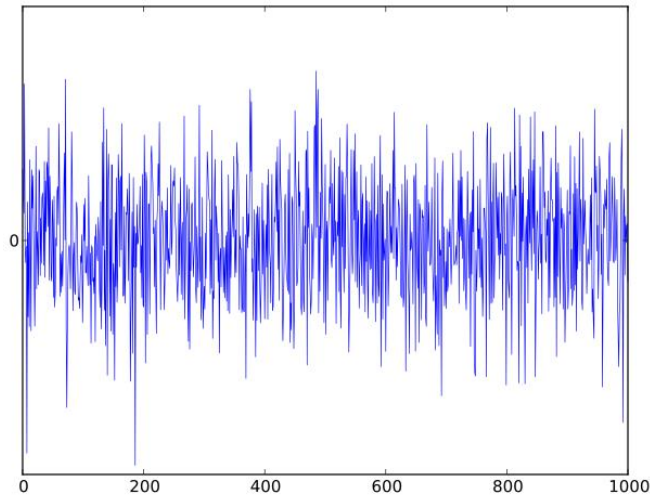
- Demo



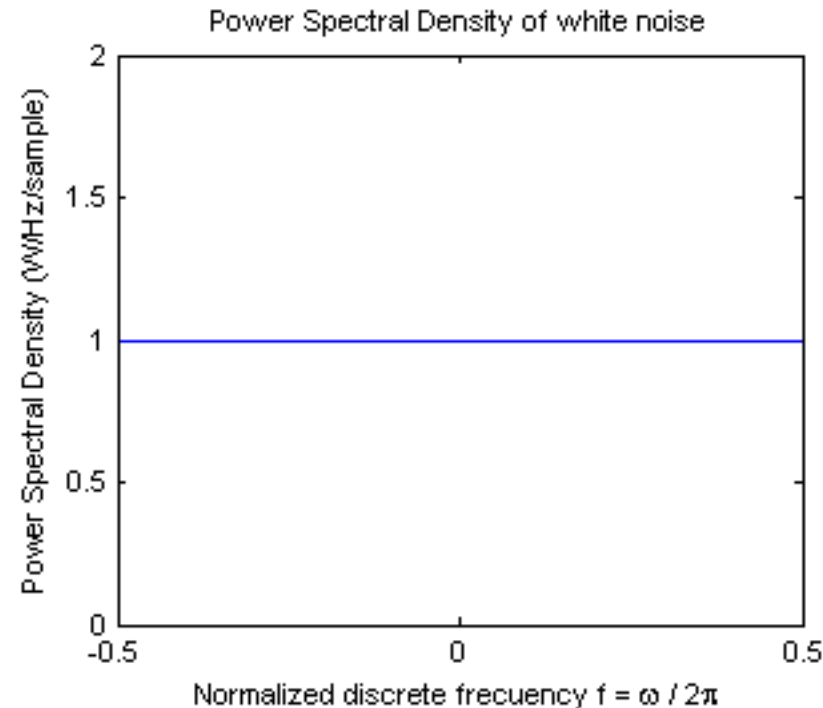
White Noise

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- In signal processing, White Noise
 - Random signal having equal intensity at different frequencies
 - Giving it a constant power spectral density.



Plot of a Gaussian white noise signal



Some "white noise" sound



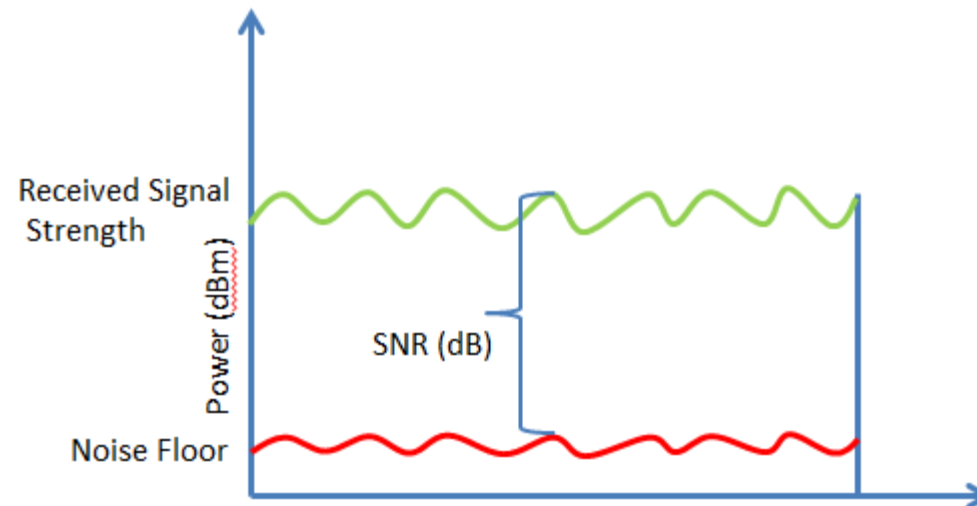
Signal to Noise Ratio

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- A signal-to-noise ratio (SNR) is a measure of signal strength relative to background noise measured in decibels (dB)

$$SNR = 20 \log_{10} \frac{S_{(volt)}}{N_{(volt)}}$$

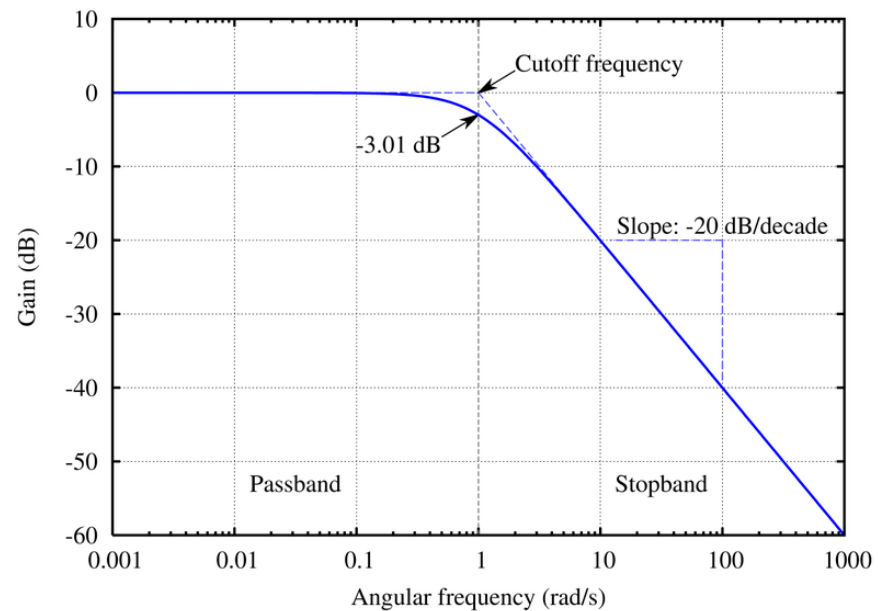
$$SNR = 10 \log_{10} \frac{S_{(watt)}}{N_{(watt)}}$$



Filtering

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- Low Pass Filtering
 - passes low-frequency signals and blocks high-frequency signals.

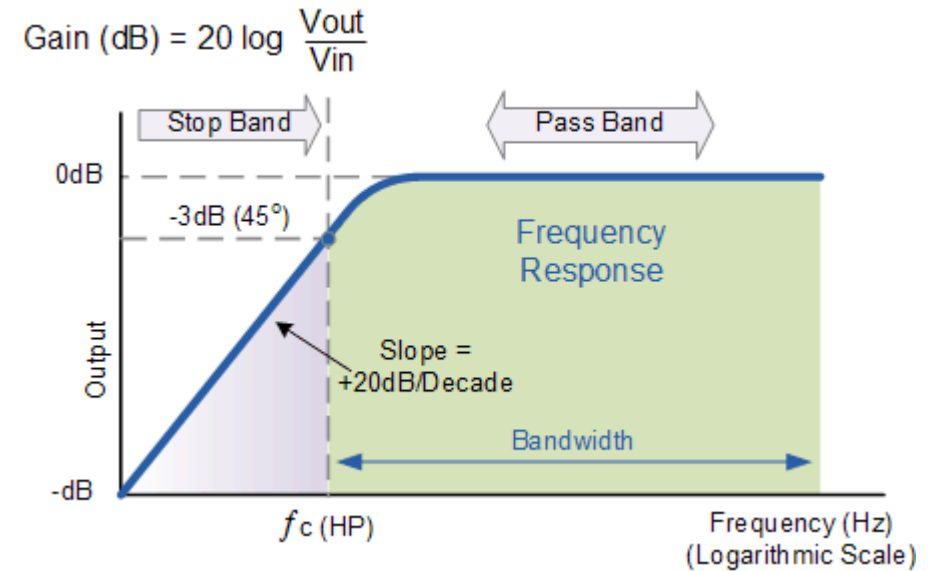


Filtering

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- High Pass Filtering

- Passes signals with a frequency higher than a certain cutoff frequency and attenuates signals with frequencies lower than the cutoff frequency

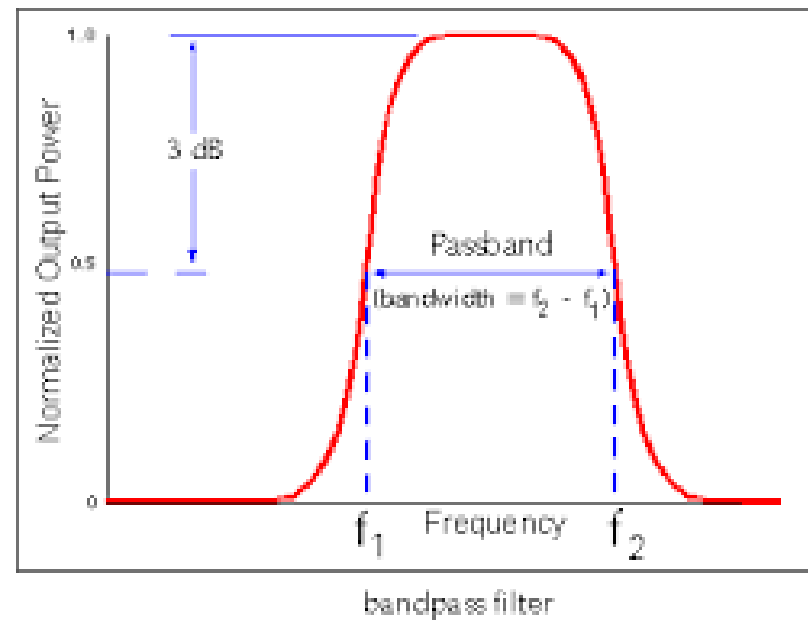


Filtering

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- Band Pass Filtering

- allows signals between two specific frequencies to pass
- rejects (attenuates) frequencies outside that range

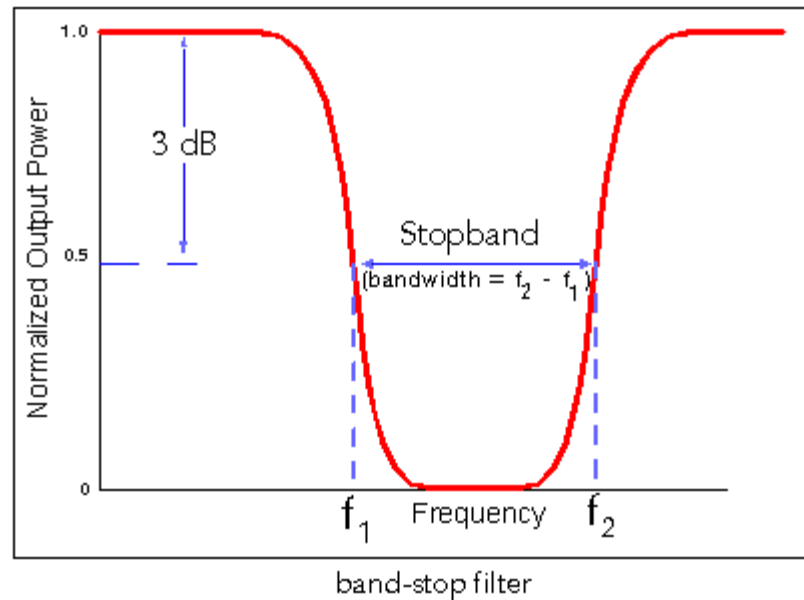


Filtering

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- **Band Stop Filtering**

- Rejects signals between two specific frequencies to pass
- allows frequencies outside that range



- Demo Signal Filtering on Matlab

```
%True Signal
%x = linspace(-2*pi,2*pi,100);
T = 100; t = (0:T - 1);
Y = 25*sin(4 * pi * t / T);
```

```
%Noise
n=wgn(1,T,10);
%n= randn(T, 1);
```

```
%Noisy Signal
yn=Y + n;
```

```
%Filter
N = 5; b = 1 / N * ones(1, N);
yh1 = filter(b, 1, yn);
```



Demo

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- Demo Signal Filtering on Matlab

```
%Plot signals
```

```
a1 = plot(t,Y);
```

```
m1='True';
```

```
hold on
```

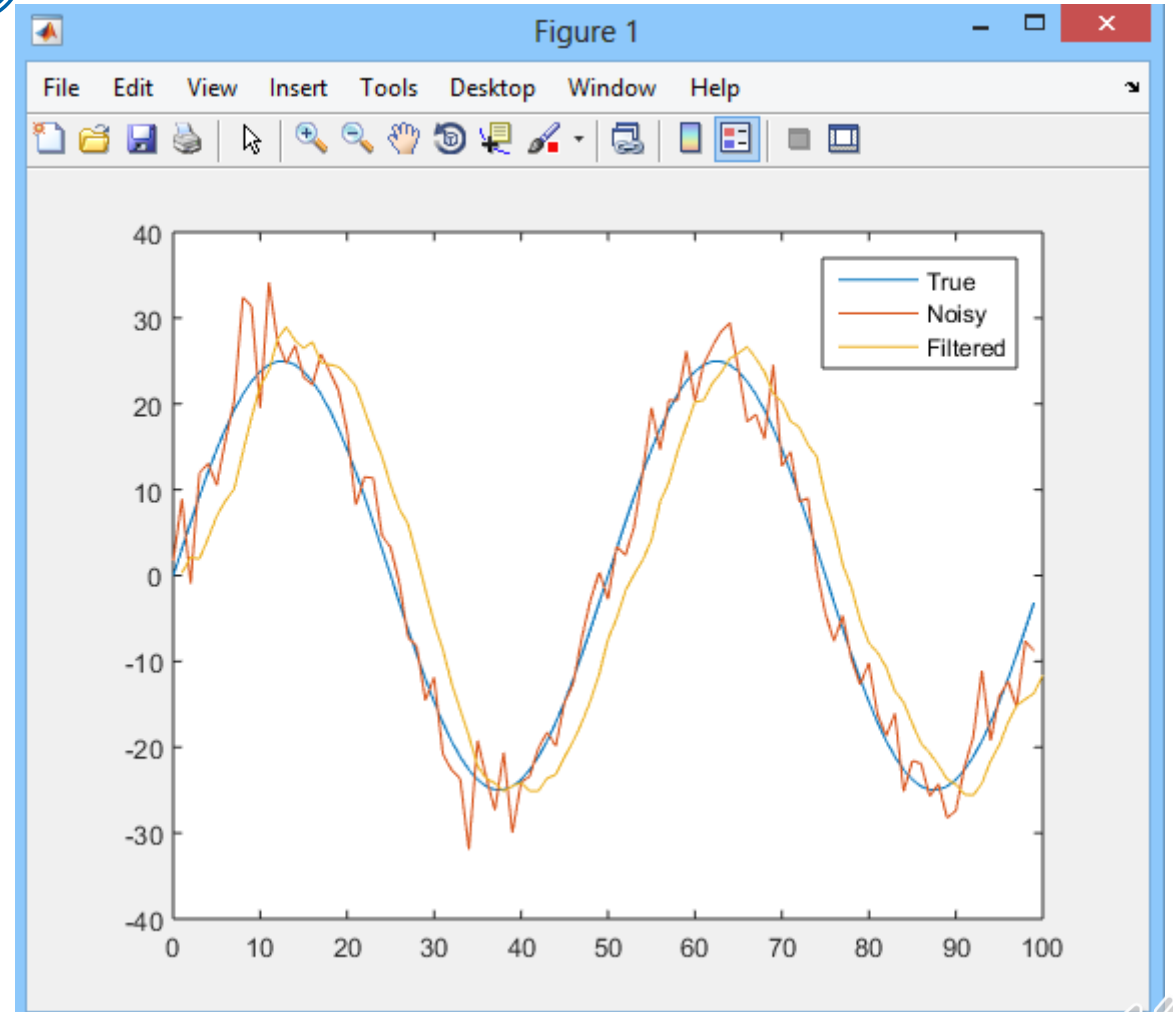
```
a2 = plot(t,yn);
```

```
m2='Noisy';
```

```
a3=plot(yh1);
```

```
m3='Filtered';
```

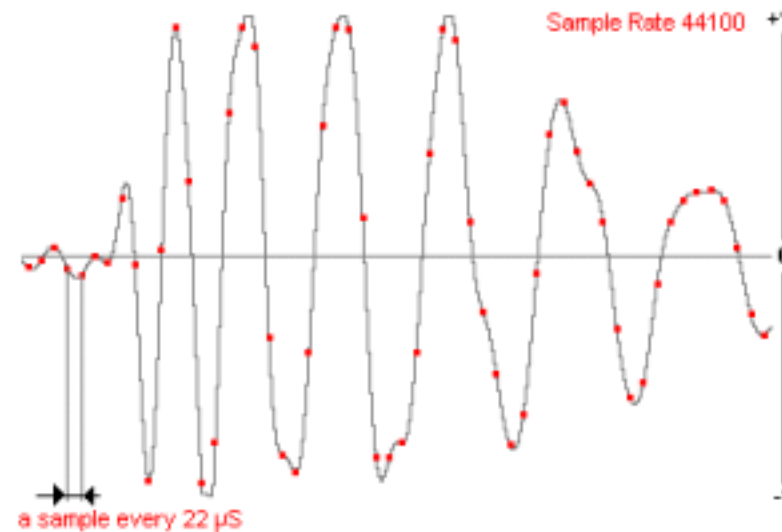
```
legend([a1;a2;a3],m1,m2,m3)
```



Sampling

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- A continuous signal $s(t)$ can be sampled by an analog-to-digital converter (A/D) to become a discrete signal composed of a set of time samples $s(nt)$
- The sampling rate (SR) is the number of times a signal is read in a second.



Sampling Theorem

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- Let $s(t), t \in \mathbb{R}$, be a band-limited signal according to

$$S(f) = 0 \quad \text{for } |f| > B$$

- If the sampling frequency F_c is at least twice the band, $F_c \geq 2B$, then $s(t)$ can be recovered by its samples $s(nt), n \in \mathbb{Z}$, according to the reconstruction formula

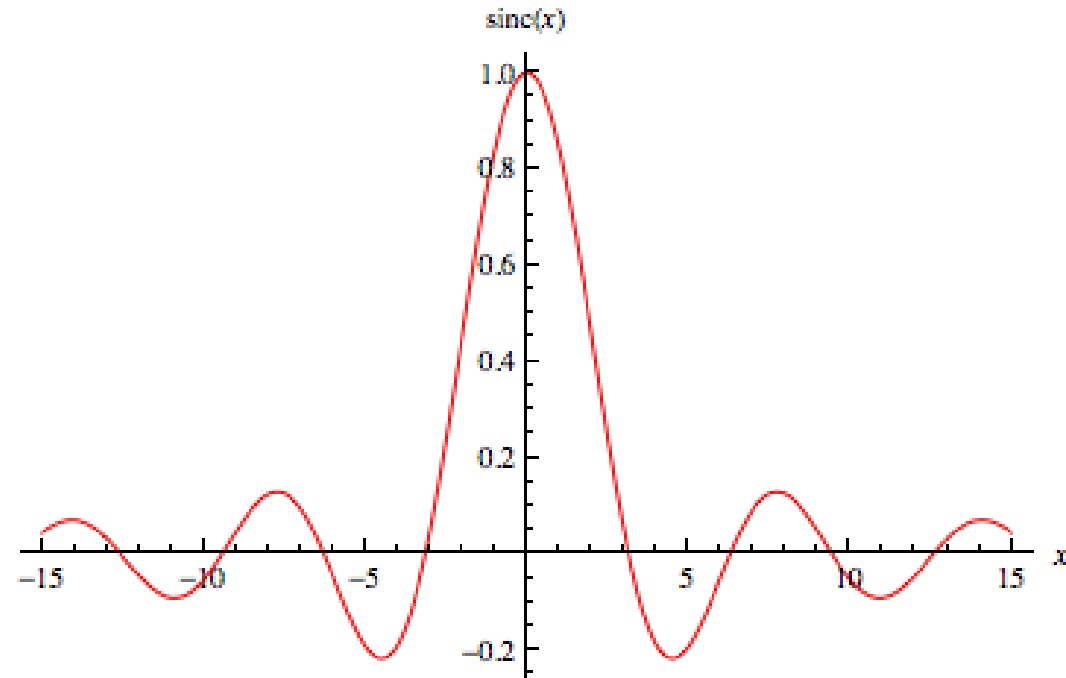
$$s(t) = \sum_{-\infty}^{+\infty} s(nt) \operatorname{sinc}[F_c(t - nT)]$$



Sampling Theorm

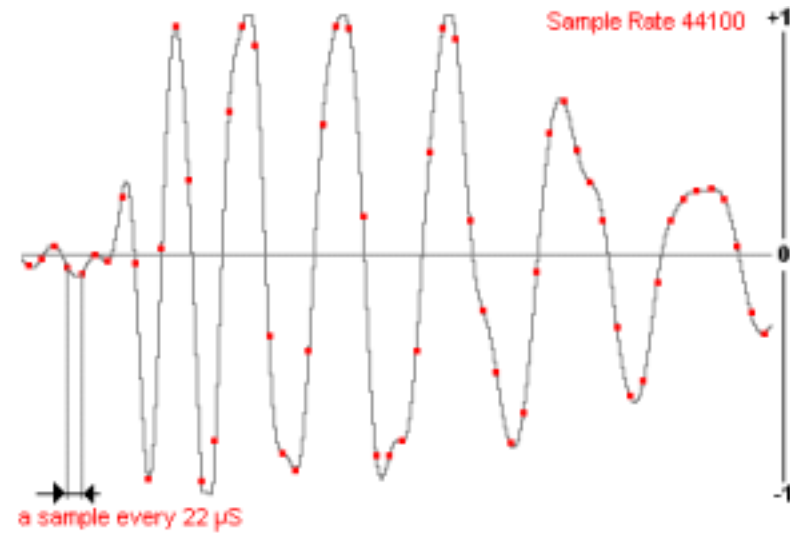
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- $\text{sinc}(x) = \begin{cases} 1 & \text{for } x = 0 \\ \frac{\sin x}{x} & \text{otherwise,} \end{cases}$



Sampling

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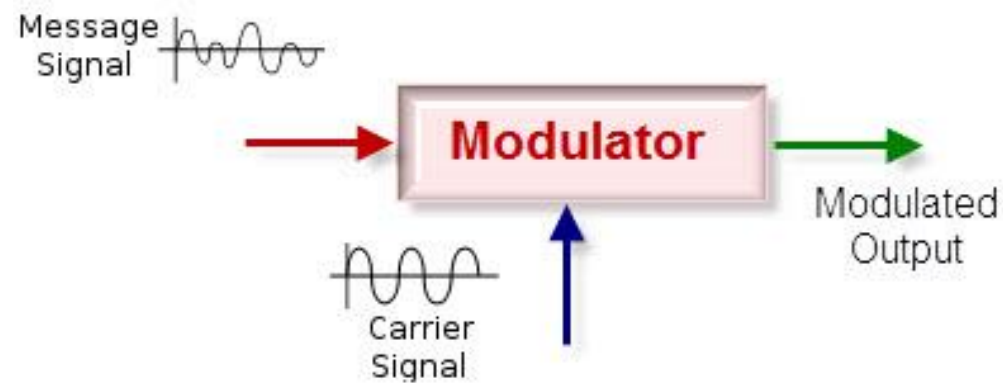


Modulation

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- **Modulation**

- Addition of information to an electronic or optical carrier signal.
- A carrier signal is one with a steady waveform -- constant amplitude and frequency.
- Information can be added to the carrier by varying its amplitude, frequency, phase, polarization (for optical signals).



- Need for Modulation

- Transmit the information to long distance without interference.
- To separate signal from different transmitters :
 - ✦ Audio frequencies are within the range of 20 Hz to 20 kHz. Without modulation all signals at same frequencies from different transmitters would be mixed up
- Size of the antenna:
 - ✦ For efficient transmission the transmitting antennas should have length at least equal to a quarter of the wavelength of the signal to be transmitted.
 - ✦ For an electromagnetic wave of frequency 15 kHz, the wavelength λ is 20 km and one-quarter of this will be equal to 5 km. Obviously, a vertical antenna of this size is impracticable. On the other hand, for a frequency of 1 MHz, this height is reduced to 75 m.



Modulation

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- Amplitude Modulation AM

- The amplitude of the carrier signal varies in accordance with the message signal, and other factors like phase and frequency remain constant.

- Modulating signal:

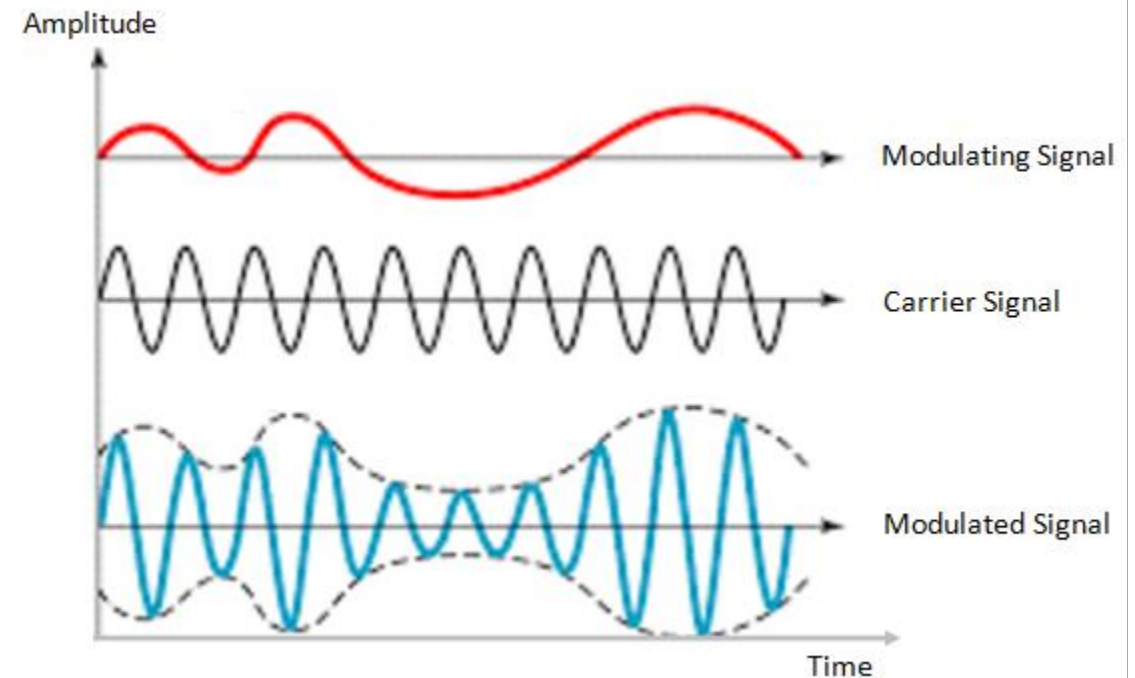
$$m(t) = A_m \cos(2\pi f_m t)$$

- Carrier signal:

$$c(t) = A_c \cos(2\pi f_c t)$$

- Amplitude Modulated wave:

$$s(t) = [A_c + A_m \cos(2\pi f_m t)] \cos(2\pi f_c t)$$

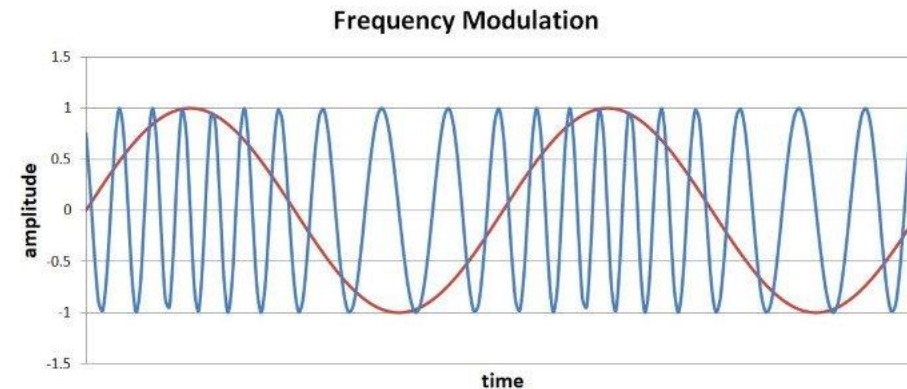
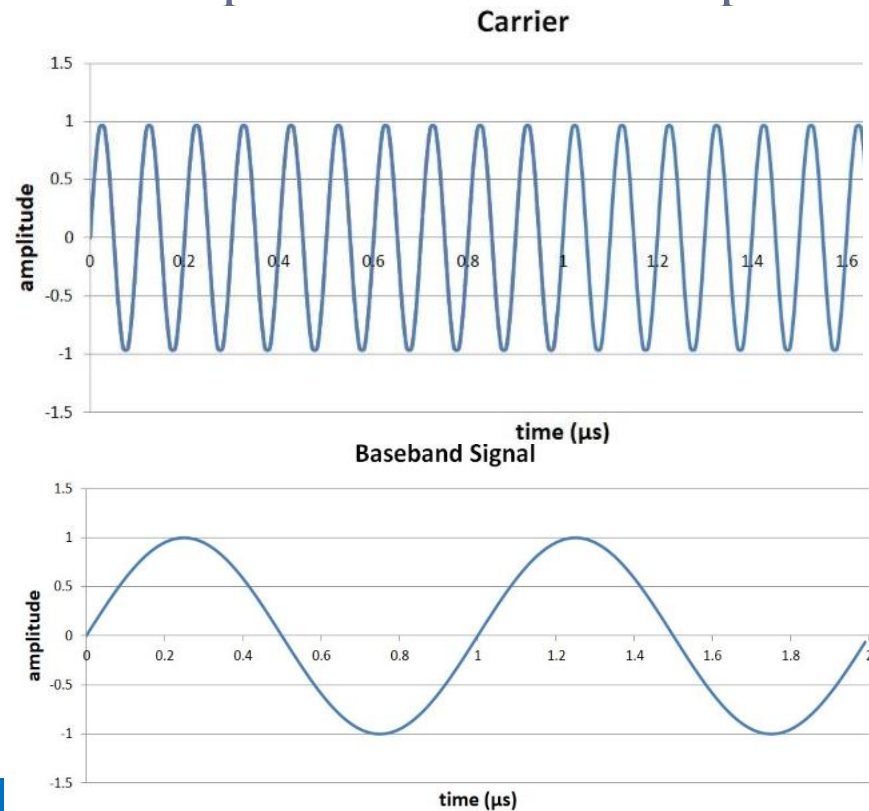


Modulation

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- Frequency Modulation FM

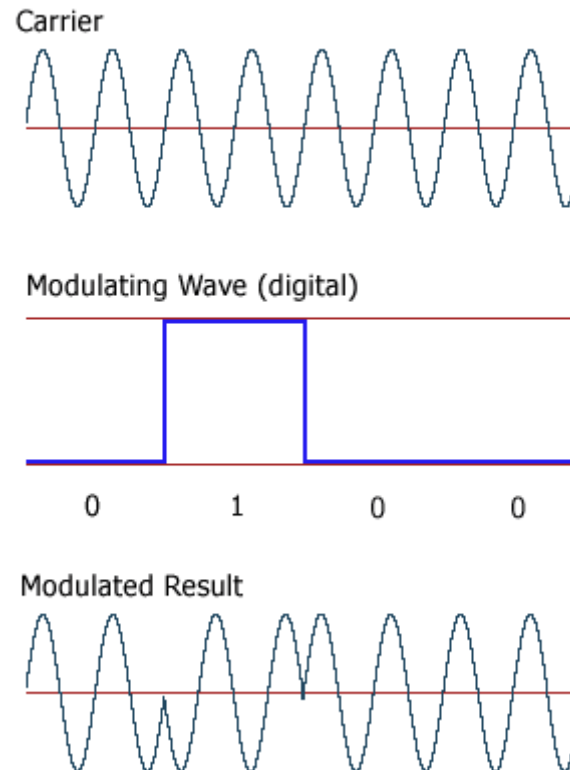
- The frequency of the carrier signal varies in accordance with the message signal, and other parameters like amplitude and phase remain constant



Modulation

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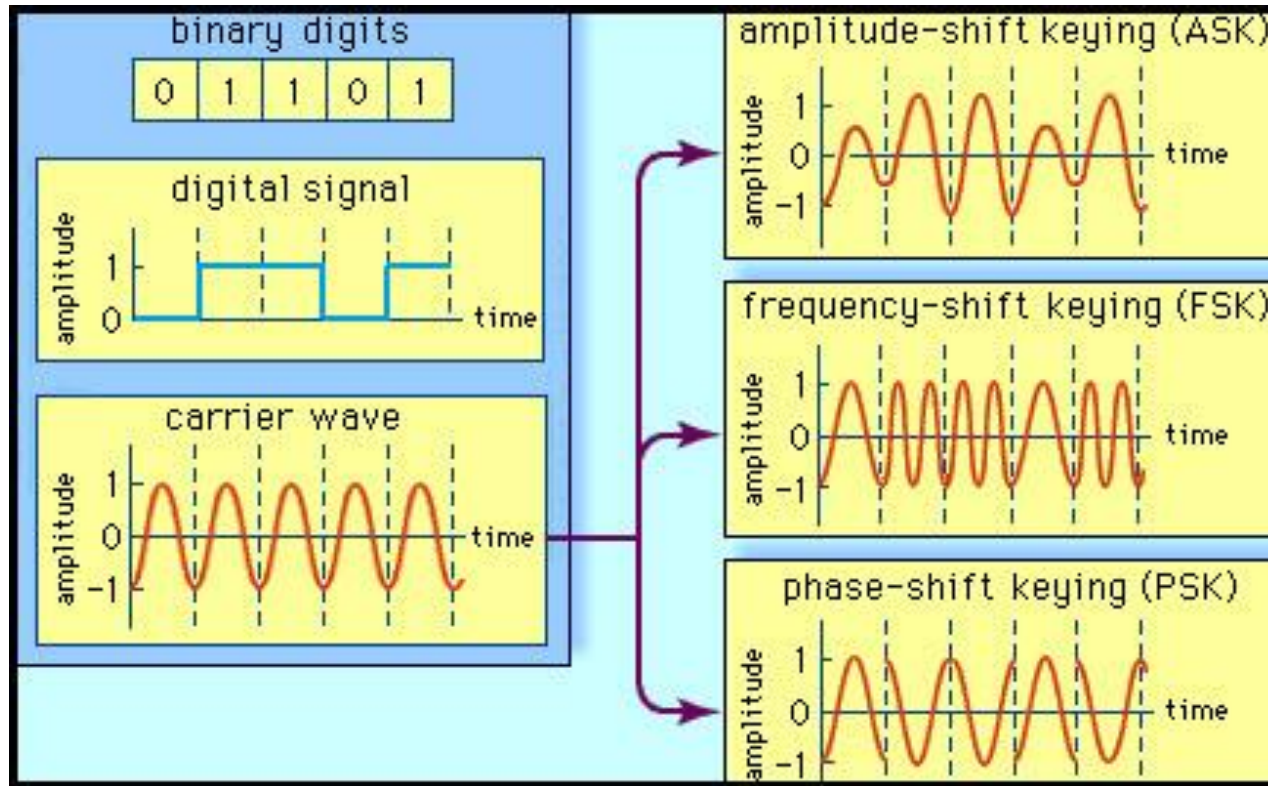
- Phase Modulation PM
 - the phase of the carrier signal varies in accordance with the message signal.



Digital Modulation

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- ASK, FSK, PSK



THANK YOU!

